

Roll No.....

EVEN END SEMESTER EXAMINATION JUNE-2025

Name of the Course: B. Tech.

Semester: II

Name of the Paper: Engineering Physics

Paper Code: TPH 201

Time: 3 Hours

Maximum Marks: 100

- Note:
- (i) All questions are compulsory
 - (ii) Answer any two sub questions among a, b, & c in each main question
 - (iii) Total marks in each main question are twenty.
 - (iv) Each sub question carries 10 marks.

Q1	(20 marks)	
(a)	Explain why Newton's rings formed are of circular shape. Discuss the effect of introducing liquid (μ) on the original diameter of the rings (with air) derive the necessary expression to compare the diameters of rings in liquid and air.	CO 1
(b)	In N slits Fraunhofer diffraction deduce the ratio of intensity of principal maxima to the intensity of secondary maxima. Discuss the effect on intensity pattern when the number of slits N is increased.	
(c)	Derive the expression for resolving power of grating. What is the highest order spectra which will be seen with light of wavelength 5893Å by means of grating with 5000lines/cm?	
Q2	(20 marks)	
(a)	Derive the relation between Einstein coefficients for the spontaneous emission to stimulated emission.	CO 2
(b)	Discuss the method for production of elliptically and circularly polarized light. Also derive the necessary condition to show that circularly polarized light is one special case of elliptically polarized light.	
(c)	(a) The linearly polarized light of intensity I_0 is incident on combination of two Nicols N_1 and N_2 . If two Nicols are first crossed (placed perpendicular) and then one of them is rotated through 60° , then find the intensity of light transmitted from N_2 . (b) Define specific rotation. Discuss the construction and working of Laurent's Half Shade Polarimeter.	
Q3	(20 marks)	
(a)	Discuss the negative results of Michelson Morley experiment. Derive Einstein mass energy relation.	CO 3
(b)	Write the two postulates for special theory of relativity. Derive the expression for time dilation; give example to show that it is a real effect.	
(c)	What is the length of one meter stick moving parallel to its length, when its mass is $3/2$ times its rest mass.	
Q4	(20 marks)	
(a)	Apply Schrodinger wave equation for a particle in a one box of infinite potential and show that the energy levels are quantized, also derive the normalized eigen wave function for this particle.	CO4 & CO5
(b)	Explain the physical significance of wave function Ψ and write its properties. Calculate the uncertainty in measurement of momentum of an electron if the uncertainty in location it is 1Å.	
(c)	Write short notes on Quantum gates, Quantum parallelism and quantum interference.	
Q5	(20 marks)	
(a)	Derive the London equation and discuss how it explains Meissner effect.	CO5 & CO6
(b)	Derive four Maxwell's equation and write their physical significance.	
(c)	Write short notes on various techniques used for the synthesis of nanoparticles; also discuss chemical vapor deposition (CVD) in details.	

